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Real-Time Big Data Ingestion Pipeline from Distributed Sources

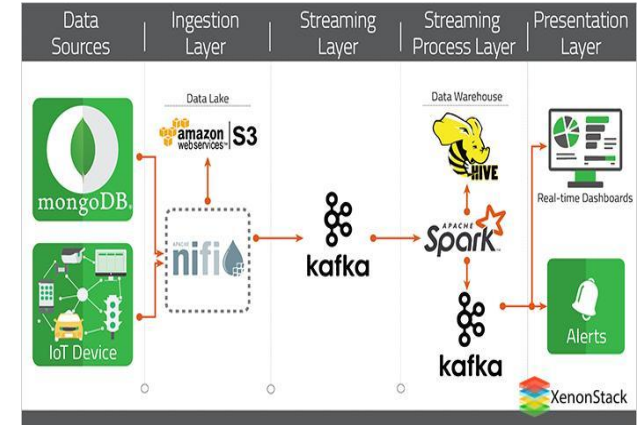
CSA1522-CLOUD COMPUTING & BIG DATA ANALYTICS

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Introduction

- Big Data applications generate massive volumes of data from multiple distributed sources in real time.
- Data may come from IoT devices, sensors, web applications, social media, servers, and enterprise systems.
- Traditional data processing methods may not handle high-velocity and continuously generated data efficiently.
- A real-time big data ingestion pipeline collects, processes, stores, and visualizes data with minimum delay.



Abstract

- Ingests high-velocity data from multiple sources.
- Uses **Apache Kafka** for real-time data ingestion.
- Uses **Apache Spark Streaming** for data processing.
- Uses **HDFS/S3** for large-scale data storage.
- Uses **Grafana/Kibana** for real-time visualization.
- Provides reliable and scalable data processing.

Literature Review

S.NO	Journal/Reference	Algorithm/Technology	Advantages	Disadvantages	Research Gap
1	Raptis et al., 2022 – Design Guidelines for Apache Kafka Driven Data Management and Distribution in Smart Cities	Apache Kafka, Apache NiFi, Distributed Data Management, Event Streaming	Provides a modular architecture for data acquisition, management, distribution, and processing. Supports scalable handling of complex data flows from sensors and devices. (arXiv)	Mainly focuses on smart-city applications and requires careful configuration for different data-flow requirements.	Limited focus on a general-purpose ingestion pipeline combining multiple distributed sources such as databases, APIs, IoT devices, and application logs.
2	A Scalable Framework for Data Lakes Ingestion, 2022 – Procedia Computer Science	Apache NiFi, Data Lake, Distributed Data Ingestion, ETL	Provides a scalable and robust architecture for ingesting heterogeneous data from multiple sources. The study also evaluates the architecture using benchmarking experiments. (ScienceDirect)	Primarily targets data-lake ingestion and may require additional streaming technologies for extremely low-latency applications.	Need for stronger integration of real-time streaming, dynamic load balancing, fault tolerance, and continuous monitoring in a unified ingestion pipeline.
3	Duraisamy Rajamani et al., 2026 – Kafka-Driven Scalable Streaming Pipelines for Real-Time Sensor Ingestion and High-Throughput Data Lakehouse Architecture	Apache Kafka, Distributed Stream Processing, Data Lakehouse, Real-Time Sensor Ingestion	Addresses high-velocity sensor data using Kafka and distributed processing with a scalable data-lakehouse architecture. It is specifically designed for real-time ingestion and high-throughput environments. (JIEM)	Focuses mainly on sensor/industrial streaming data and does not fully cover all types of distributed enterprise sources.	Further research is needed for a unified multi-source pipeline supporting IoT, databases, APIs, logs, heterogeneous formats, preprocessing, validation, and reliable delivery.

Problem statements

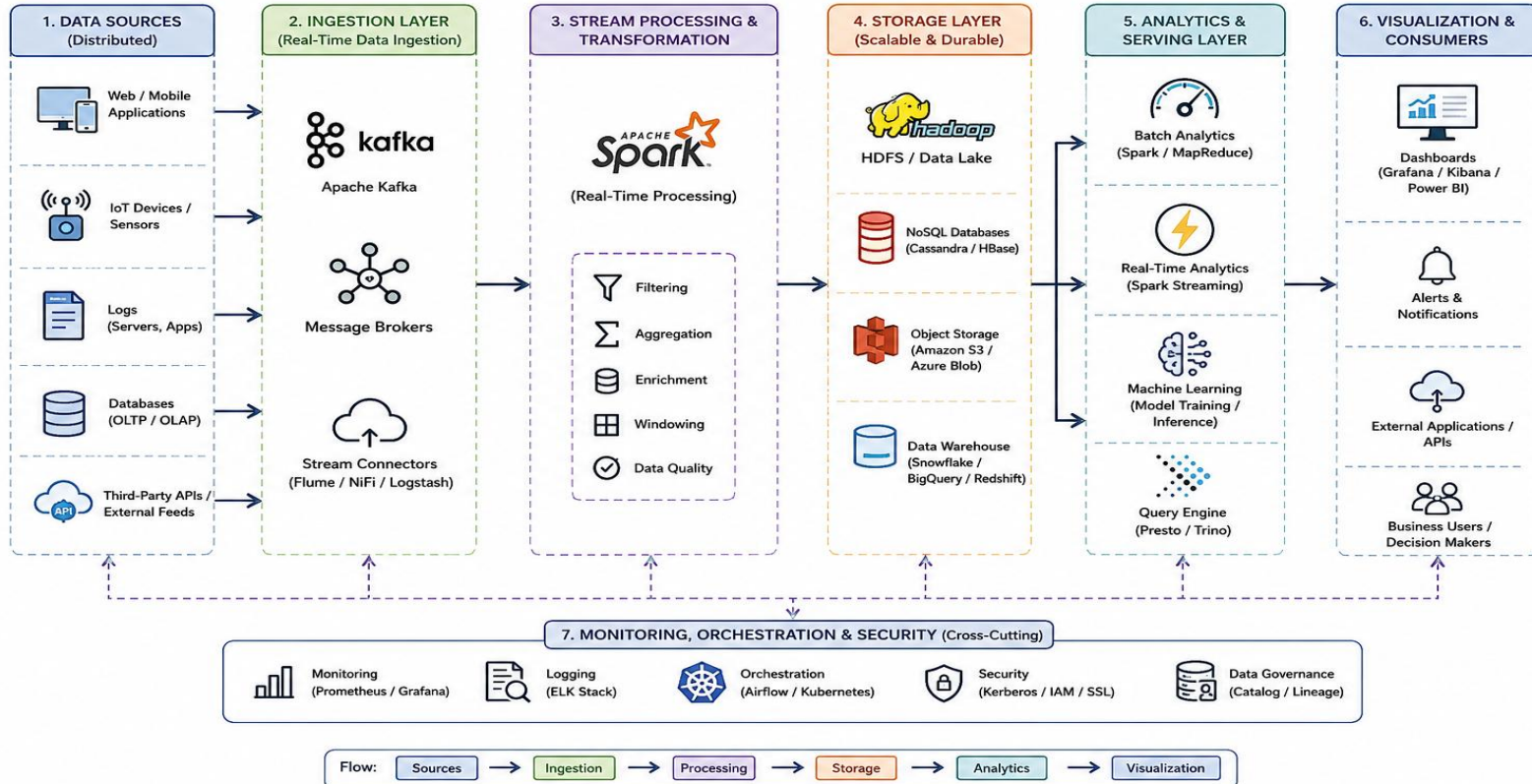
- Modern applications generate huge volumes of data continuously from distributed sources such as IoT devices, applications, APIs, databases, and system logs.
- Traditional batch-processing systems cannot process this continuously generated data with low latency.
- Data arrives at different speeds and in different formats, making integration and processing difficult.
- Delayed data processing can affect real-time monitoring and decision-making.

Objectives

- To design a real-time pipeline for ingesting high-velocity data from multiple distributed sources.
- To use Apache Kafka for reliable distributed messaging and real-time data ingestion.
- To process and transform streaming data using Apache Spark Streaming.
- To store large-scale processed data using HDFS or cloud storage such as Amazon S3.
- To visualize real-time analytics and pipeline performance using Grafana/Kibana.

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System Architecture

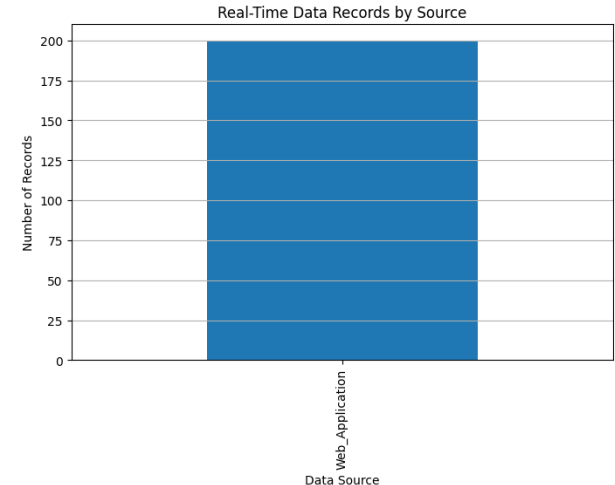


Module 1 – Multi-Source Data Ingestion Module

- Collects real-time data from multiple distributed sources such as IoT devices, sensors, applications, and server logs.
- **Apache Kafka** acts as the distributed messaging and data ingestion platform.
- Kafka topics organize incoming data streams and allow multiple consumers to process data independently.

Features

- Multi-Source Data Collection
- Apache Kafka Producers
- Kafka Topics and Partitions

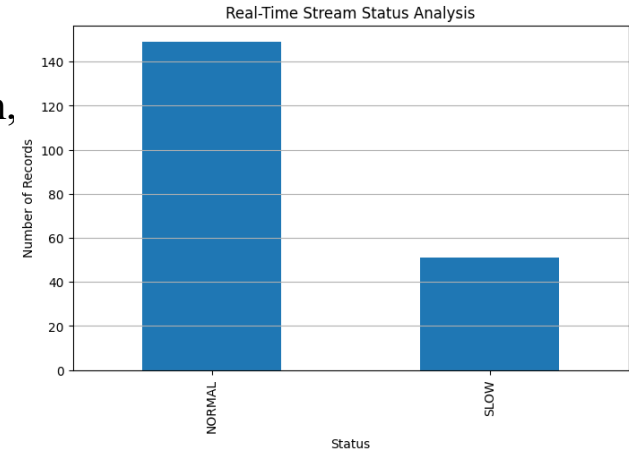


Module 2 – Stream Processing & Transformation Engine

- Receives continuous data streams from Apache Kafka.
- **Apache Spark Streaming** processes incoming data in real time or near real time.
- Performs data cleaning, filtering, transformation, aggregation, and feature extraction.

Features

- Real-Time Stream Processing
- Data Cleaning
- Data Filtering
- Data Transformation

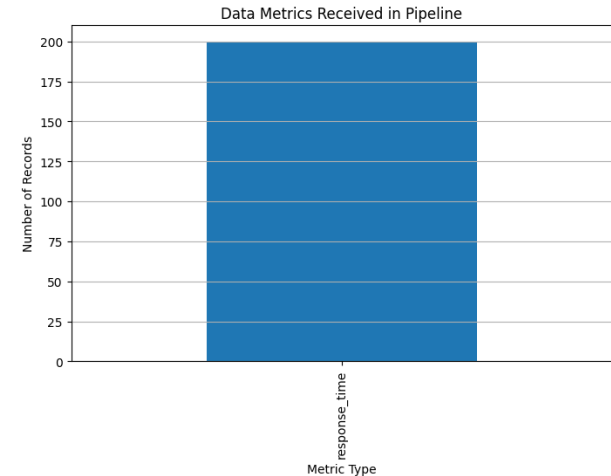


Module 3 – Cloud Storage & Visualization Dashboard

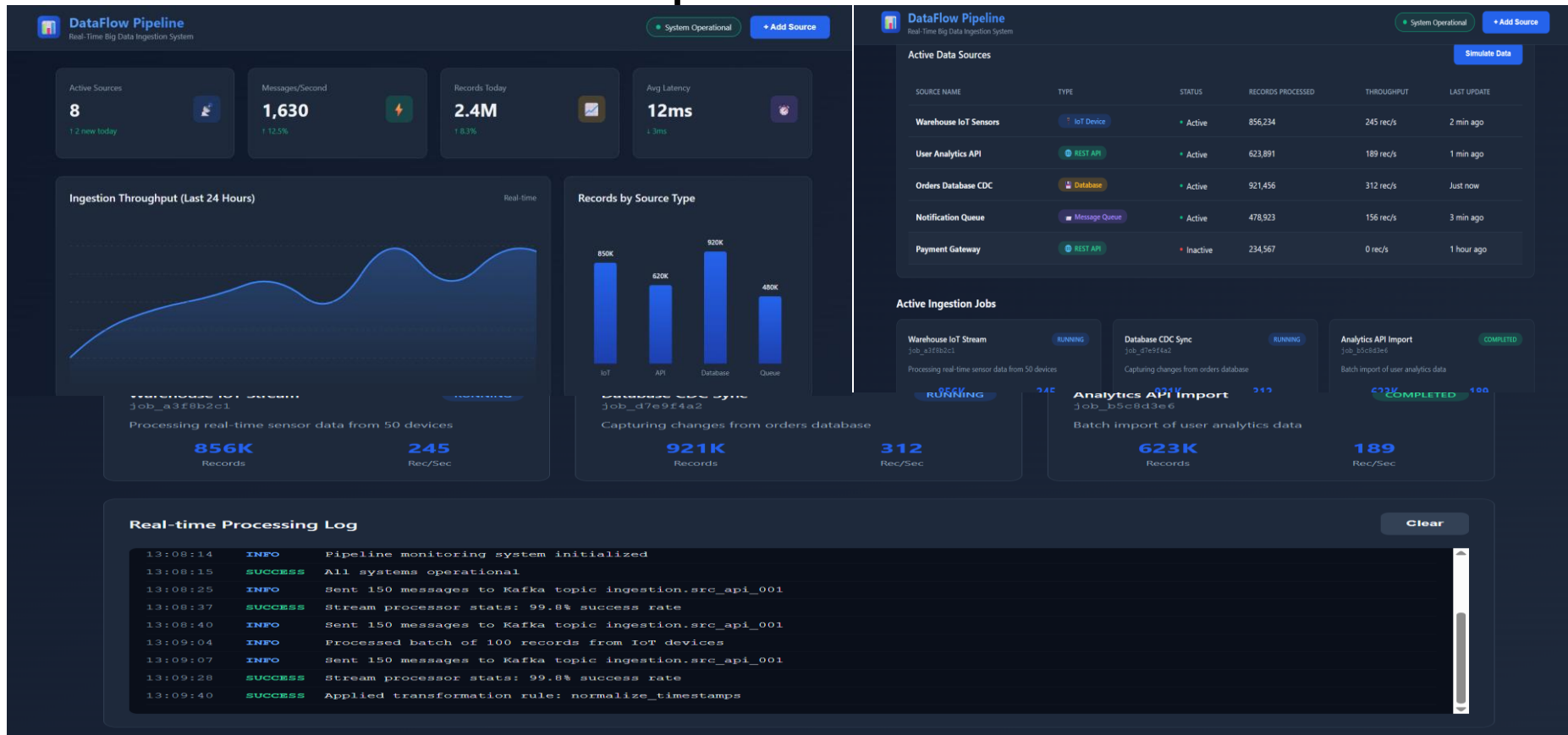
- Stores raw and processed big data in distributed storage systems.
- **HDFS** can be used for distributed on-premise storage, while **Amazon S3** provides scalable cloud-based object storage.
- Processed data is made available for analytics and long-term storage.

Features

- HDFS/S3 Data Storage
- Scalable Cloud Storage
- Historical Data Management
- Real-Time Data Visualization



Implementation



Techniques Used

- Stream Processing
- Distributed Messaging using Apache Kafka
- Big Data Pipeline Design
- Data Serialization
- Apache Spark Streaming
- Distributed Storage using HDFS/S3
- Real-Time Data Visualization
- 5Vs of Big Data: Volume, Velocity, Variety, Veracity, and Value

Applications

- IoT and Smart City Data Processing
- Real-Time Financial Transaction Monitoring
- Fraud Detection Systems
- Social Media Analytics
- Healthcare Monitoring Systems
- Industrial and Manufacturing Monitoring
- E-Commerce Recommendation Systems



Future Scope

- Integrate more IoT and enterprise data sources.
- Implement advanced anomaly detection.
- Add machine-learning-based real-time prediction.
- Use containerization with Docker and Kubernetes.
- Implement automatic scaling.

Conclusion

- Provides real-time ingestion of distributed data.
- Kafka enables reliable data streaming.
- Spark Streaming enables fast data processing.
- HDFS/S3 provides scalable data storage.
- Grafana/Kibana provides real-time visualization.
- The pipeline supports reliable and low-latency analytics.

References

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**THANK
YOU**